American Terawatt

Fervo Cape Station

PSLF model representation — physical network and dynamic controls

Revision history

REV.	DATE	AUTHOR	DESCRIPTION
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AMERICA'S PRIVATE INDUSTRIAL POWER GRID

1 Scope

This document sets out the PSLF model of Fervo Cape Station Phase 2 as delivered in the two source files listed in Section 10: a power-flow network case (`.epc`, PSLF version 24.3.0) [1] and a dynamic-data file (`.dyd`) [2]. The representation is split into a *physical part* — buses, lines, transformers, generators and loads that define the steady-state network — and a *control part* — the machine, excitation, governor, stabilizer and protection models that define the dynamic behaviour of each unit.

The plant comprises nine steam-turbine generators at 13.8 kV, arranged in pairs on four generator buses, each pair stepped up through its own 150 MVA generator step-up transformer (GSU) to a 345 kV collector bus (GEN-TIE). The collector bus connects over a 3.0 mile line to the point of interconnection (POI), which is represented as an infinite bus in the dynamic data. A 3.2 mile line from GEN-TIE to the Phase 1 main power transformer bus (HV-MPT-P1) is included in the network case as the tie to Phase 1.

Two families of units appear in the data. Units 4, 5 and 6 are Turboden machines rated 76.471 MV·A, and units 7 through 11 are Baker Hughes machines rated 71.13 MV·A. The two families share the same excitation, stabilizer and protection settings but have different machine constants and turbine capabilities. Every parameter value in the tables of this document is transcribed from the source files by script, so the tables can be regenerated whenever the files change.

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2 Physical network model

The network case contains 11 buses, 6 branches, 4 transformers, 9 generators plus the POI swing source, and 4 loads. All equipment is assigned to area 26 (LADWP) and zone 283 (LA IPP). The system base is 100 MV·A. Figure 1 shows the one-line arrangement.

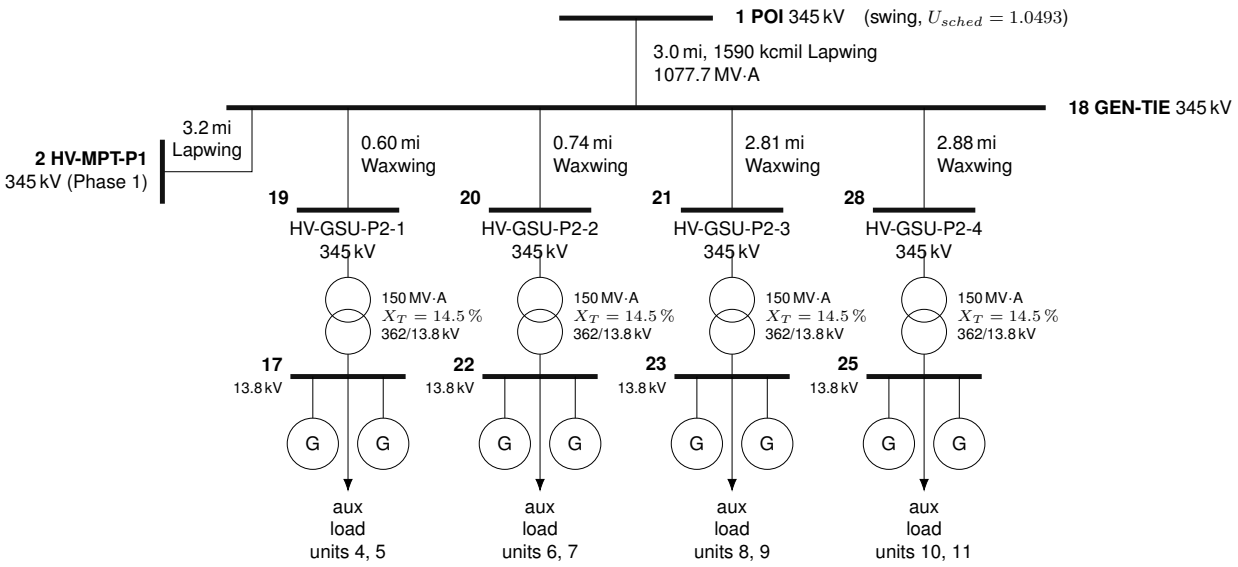


Figure 1: One-line diagram of the Phase 2 network case. Bus numbers are those of the `.epc` file; line lengths and conductor types are taken from the branch records.

2.1 Buses

Table 1 lists the buses. Bus 1 is the swing bus (PSLF type 0) with a scheduled voltage of 1.0493 pu; the four generator buses are voltage-controlled (type 2) at 1.00 pu, regulating their own terminals; the remaining 345 kV buses are load buses (type 1). Every bus carries the same 0.95–1.05 pu voltage band.

No.	Name	kV	Type	U_{sched}	Function
1	POI	345.0	0	1.0493	Point of interconnection, swing bus
2	HV-MPT-P1	345.0	1	–	Phase 1 main power transformer HV bus
18	GEN-TIE	345.0	1	–	Phase 2 345 kV collector bus
19	HV-GSU-P2-1	345.0	1	–	GSU 1 HV terminal
20	HV-GSU-P2-2	345.0	1	–	GSU 2 HV terminal
21	HV-GSU-P2-3	345.0	1	–	GSU 3 HV terminal
28	HV-GSU-P2-4	345.0	1	–	GSU 4 HV terminal
17	GEN-P2-U4-5	13.8	2	1.0000	Generator bus, units 4 and 5, aux load
22	GEN-P2-U6-7	13.8	2	1.0000	Generator bus, units 6 and 7, aux load
23	GEN-P2-U8-9	13.8	2	1.0000	Generator bus, units 8 and 9, aux load
25	GEN-P2-U10-1	13.8	2	1.0000	Generator bus, units 10 and 11, aux load

Table 1: Bus data (all buses in area 26 LADWP, zone 283 LA IPP).

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2.2 Branches

The six 345 kV branches are given in Table 2. Impedances are in per unit on the 100 MV·A system base; `charge` is the total line charging B. The POI and Phase 1 ties use 1590 kcmil Lapwing ACSR; the four GSU feeders use 266.8 kcmil Waxwing ACSR. The GEN-TIE to HV-MPT-P1 branch carries a metering flag at the GEN-TIE end and a 90 line-section identifier in the record, which is how the Phase 1/Phase 2 boundary is marked in the case.

From	To	R	X	B	Rate 1	Rate 2	Rate 3	Rate 4	mi
1 POI	18 GEN-TIE	0.000145	0.001841	0.021185	1077.7	1291.2	1351.8	1408.4	3.00
18 GEN-TIE	2 HV-MPT-P1	0.000155	0.001963	0.022597	1077.7	1291.2	1351.8	1408.4	3.20
18 GEN-TIE	19 HV-GSU-P2-1	0.000159	0.000416	0.003687	144.6	190.9	204.0	216.2	0.60
18 GEN-TIE	20 HV-GSU-P2-2	0.000196	0.000513	0.004548	144.6	190.9	204.0	216.2	0.74
18 GEN-TIE	21 HV-GSU-P2-3	0.000745	0.001949	0.017269	144.6	190.9	204.0	216.2	2.81
18 GEN-TIE	28 HV-GSU-P2-4	0.000763	0.001997	0.017699	144.6	190.9	204.0	216.2	2.88

Table 2: Branch data (pu on 100 MV·A; ratings in MV·A; the two 3-mile ties are Lapwing, the four feeders Waxwing).

2.3 Transformers

All four GSUs are identical two-winding units, Table 3. The impedance is entered on the transformer base of 150 MV·A, so on the 100 MV·A system base it becomes

$$X_{T,sys} = 0.14496 \cdot \frac{100}{150} = 0.09664\,\text{pu}, \qquad R_{T,sys} = 0.00345 \cdot \frac{100}{150} = 0.00230\,\text{pu}. \tag{1}$$

The HV winding tap is 362 kV against a 345 kV bus base, i.e. a fixed $362/345 = 1.0493$ pu tap position, which matches the scheduled POI voltage. The transformers are of PSLF type 1 (fixed tap, no regulation) with a 200 MV·A rating in both rate slots.

HV bus	LV bus	ck	S_{base}	R_T	X_T	Tap HV/LV	Rating
			MV·A	pu	pu	kV	MV·A
19 HV-GSU-P2-1	17 GEN-P2-U4-5	1	150	0.00345	0.14496	362.0 / 13.8	200
20 HV-GSU-P2-2	22 GEN-P2-U6-7	1	150	0.00345	0.14496	362.0 / 13.8	200
21 HV-GSU-P2-3	23 GEN-P2-U8-9	11	150	0.00345	0.14496	362.0 / 13.8	200
28 HV-GSU-P2-4	25 GEN-P2-U10-1	1	150	0.00345	0.14496	362.0 / 13.8	200

Table 3: Generator step-up transformer data.

2.4 Generators

Table 4 gives the generator dispatch and limits. Each unit regulates its own terminal bus at 1.00 pu with a full share of the reactive requirement ($\texttt{prf} = \texttt{qrf} = 1.0$). The dynamic source impedance `gen_x` is 1.0 pu on the machine base for every unit and is overridden by the machine model at initialisation. The POI source is a 99 999 MV·A swing generator with $\pm$9999 MW and $\pm$9999 Mvar limits, i.e. an ideal source.

2.5 Loads

Each generator bus carries a constant-power auxiliary load representing the plant's own consumption (geothermal brine pumping and balance of plant), Table 5. The loads are modelled entirely as constant P and Q; the current- and impedance-dependent components are zero.

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Bus	id	Long name	P_{gen}	P_{max}	P_{min}	Q_{max}	Q_{min}	S_{base}
17	1	Cape Phase 2 – Turboden – Gen 4	56.659	65.000	0	45.118	-36.706	76.471
17	2	Cape Phase 2 – Turboden – Gen 5	56.659	65.000	0	45.118	-36.706	76.471
22	1	Cape Phase 2 – Turboden – Gen 6	56.659	65.000	0	45.118	-36.706	76.471
22	2	Cape Phase 2 – Baker H – Gen 7	55.402	60.461	0	46.235	-38.410	71.130
23	1	Cape Phase 2 – Baker H – Gen 8	55.402	60.461	0	46.235	-38.410	71.130
23	2	Cape Phase 2 – Baker H – Gen 9	55.402	60.461	0	46.235	-38.410	71.130
25	1	Cape Phase 2 – Baker H – Gen 10	55.402	60.461	0	46.235	-38.410	71.130
25	2	Cape Phase 2 – Baker H – Gen 11	55.402	60.461	0	46.235	-38.410	71.130
1	1	POI (swing source)	0	9999	-9999	9999	-9999	99999

Table 4: Generator data (MW, Mvar and MV·A).

Bus	Name	Long name	P (MW)	Q (Mvar)
17	GEN-P2-U4-5	gen aux load	35.60	17.24
22	GEN-P2-U6-7	gen aux load	34.36	16.64
23	GEN-P2-U8-9	gen aux load	33.12	16.04
25	GEN-P2-U10-1	gen aux load	33.12	16.04
Total			136.20	65.96

Table 5: Auxiliary loads (constant power).

2.6 Dispatch summary

The scheduled dispatch of the nine units sums to

$$P_{gross} = 3 \cdot 56.659\,\mathrm{MW} + 5 \cdot 55.402\,\mathrm{MW} = 446.98\,\mathrm{MW}, \tag{2}$$

against a combined capability of $3 \cdot 65.0 + 5 \cdot 60.461 = 497.3$ MW. After the 136.20 MW of auxiliary load the net injection into the four GSUs is $446.98 - 136.20 = 310.78$ MW before transformer and line losses. The reactive capability of the fleet spans $+366.5$ to -302.2 Mvar at the terminals.

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3 Dynamic model structure

Every one of the nine units is represented by the same chain of five PSLF models, Figure 2: a `genqec` synchronous machine, an `esac8b` excitation system, a `ggov1` turbine-governor, a `pss2a` power system stabilizer and the `lhvrt`/`lhfrt` ride-through trip models. The POI is a `gencls` classical machine with $H = 999$ s on a 9999 MV·A base and $X''_d = 0.10$ pu, which is the conventional way of making the swing bus an infinite source in a PSLF dynamic study.

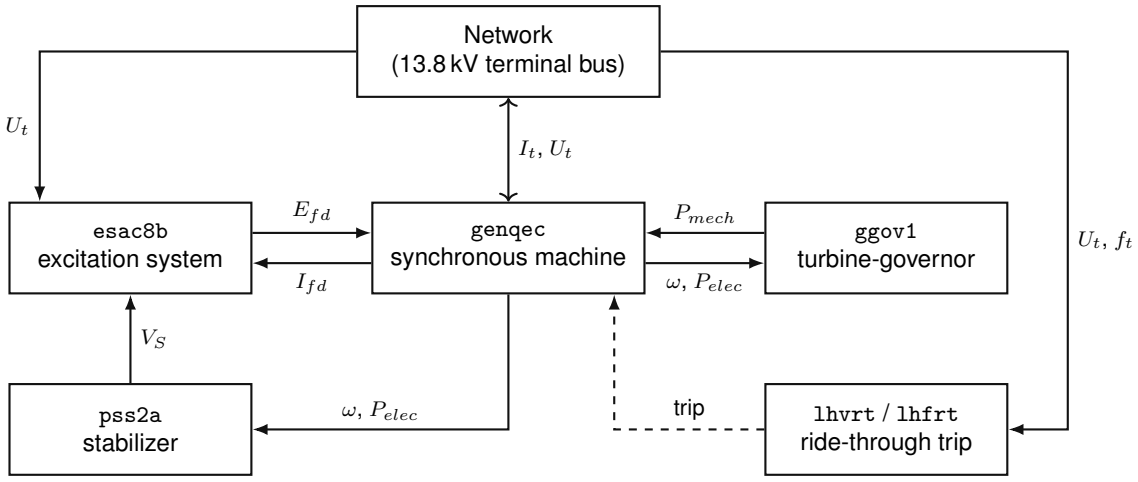


Figure 2: Signal flow between the five dynamic models attached to each unit. The stabilizer output V_S enters the exciter's voltage-error summing junction; the ride-through models trip the unit when the terminal quantities leave the permitted envelope.

The parameter sets are grouped by manufacturer. Table 6 shows which set applies to which unit; note that the two units on bus 22 belong to different families, so the two machines on that bus are not identical.

Unit	Bus	id	Machine set	Controls
Gen 4	17	1	Turboden, 76.471 MV·A, `mwcap` 65	common set
Gen 5	17	2	Turboden, 76.471 MV·A, `mwcap` 65	common set
Gen 6	22	1	Turboden, 76.471 MV·A, `mwcap` 65	common set
Gen 7	22	2	Baker Hughes, 71.13 MV·A, `mwcap` 60.461	common set
Gen 8	23	1	Baker Hughes, 71.13 MV·A, `mwcap` 60.461	common set
Gen 9	23	2	Baker Hughes, 71.13 MV·A, `mwcap` 60.461	common set
Gen 10	25	1	Baker Hughes, 71.13 MV·A, `mwcap` 60.461	common set
Gen 11	25	2	Baker Hughes, 71.13 MV·A, `mwcap` 60.461	common set

Table 6: Assignment of parameter sets to units.

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4 Synchronous machine — genqec

genqec is the round-rotor synchronous machine model that PSLF introduced as the successor to genrou and gentpj. It retains the two-axis, two-rotor-circuit-per-axis structure of genrou (transient and subtransient time constants and reactances on both axes) but represents magnetic saturation on the equivalent circuit rather than by scaling the exciting flux alone, which gives a better match to measured field current at load. The saturation function is quadratic through the points $S(1.0)$ and $S(1.2)$; kw weights the saturation between the axes and satflg selects the implementation, and both are left at their defaults of zero here. The compounding terms rcomp/xcomp are zero, so the voltage regulator sees the unloaded terminal voltage.

The two machine families differ in most electrical constants, Table 7. The Baker Hughes units have a considerably longer T'_{d0} (10 s against 5.75 s), higher synchronous and transient reactances and slightly larger inertia. For each family the ordering $L_d > L'_d > L''_d \geq L_l$ and $L_q > L'_q \geq L''_q$ holds and $T'_{d0} > T''_{d0}$, $T'_{q0} > T''_{q0}$, so the data are internally consistent. The inertia is given on the machine base; on the 100 MV·A system base the Turboden units contribute $H_{sys} = 1.49 \cdot 76.471/100 = 1.139$ s and the Baker Hughes units $H_{sys} = 1.50 \cdot 71.13/100 = 1.067$ s each.

Parameter	Description	Unit	Turboden	Baker Hughes
mva	Machine MVA base	MV·A	76.471	71.13
tpdo	d-axis transient open-circuit time constant T'_{d0}	s	5.751	10
tppdo	d-axis subtransient open-circuit time constant T''_{d0}	s	0.044	0.05
tpqo	q-axis transient open-circuit time constant T'_{q0}	s	0.865	0.865
tppqo	q-axis subtransient open-circuit time constant T''_{q0}	s	0.072	0.13
h	Inertia constant H	MW·s/MV·A	1.49	1.5
d	Damping factor D	pu	3	3
ld	d-axis synchronous reactance L_d	pu	1.436	1.49
lq	q-axis synchronous reactance L_q	pu	0.61	0.78
lpd	d-axis transient reactance L'_d	pu	0.189	0.21
lpq	q-axis transient reactance L'_q	pu	0.61	0.78
lppd	d-axis subtransient reactance L''_d	pu	0.121	0.14
lppq	q-axis subtransient reactance L''_q	pu	0.181	0.18
ll	Stator leakage reactance L_l	pu	0.0991	0.11
s1	Saturation factor at 1.0 pu flux $S(1.0)$	–	0.1	0.1
s12	Saturation factor at 1.2 pu flux $S(1.2)$	–	0.522	0.522
ra	Stator resistance R_a	pu	0.002	0.0045
rcomp	Compounding resistance (voltage-regulator compensation)	pu	0	0
xcomp	Compounding reactance (voltage-regulator compensation)	pu	0	0
kw	Saturation weighting factor K_w	–	0	0
satflg	Saturation-function selector	–	0	0

Table 7: genqec parameters (pu on machine base unless noted).

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5 Excitation system — esac8b

esac8b is the PSLF implementation of the IEEE 421.5 type AC8B excitation system: a PID voltage regulator driving a brushless AC exciter with a rotating rectifier. Figure 3 shows the block structure with the parameters used here. The regulator gains are $K_{PR} = 100$, $K_{IR} = 90$ and $K_{DR} = 20$ with $T_{DR} = 0.1$ s; the amplifier is a unity gain with $T_A = 0$, so the regulator output V_R is limited directly to $0 \le V_R \le 20$ pu. The exciter integrator has $T_E = 0.4$ s, and the feedback V_{FE} collects self-excitation $K_E \cdot V_E$, saturation $S_E(V_E) \cdot V_E$ and the armature-reaction term $K_D \cdot I_{fd}$ with $K_D = 1.19$. The rectifier regulation is $K_C = 0.17$, applied through the standard $F_{EX}(I_N)$ characteristic with $I_N = K_C \cdot I_{fd}/V_E$. The field-current limit $V_{FEMAX} = 11.46$ pu is active; the terminal-voltage and speed multipliers are off.

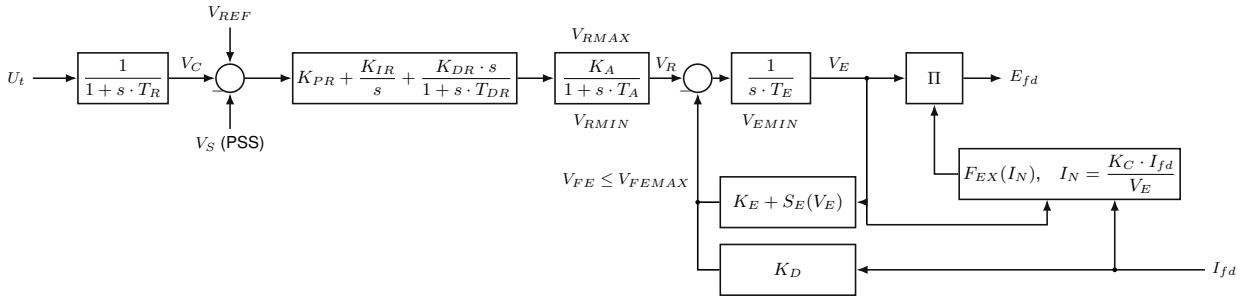


Figure 3: esac8b (IEEE AC8B) block diagram. V_S is the stabilizer output; V_{FE} is the exciter field current, limited to V_{FEMAX}.

The settings are identical for both machine families, Table 8. The saturation points $S_E(4.8) = 0.0222$ and $S_E(3.0) = 0.0219$ describe an exciter that is almost linear over its working range.

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Parameter	Description	Unit	Value
`tr`	Terminal-voltage transducer time constant T_R	s	0.01
`kpr`	Regulator proportional gain K_{PR}	pu	100
`kir`	Regulator integral gain K_{IR}	pu/s	90
`kdr`	Regulator derivative gain K_{DR}	pu·s	20
`tdr`	Derivative-block time constant T_{DR}	s	0.1
`vrmax`	Regulator output upper limit V_{RMAX}	pu	20
`vrmin`	Regulator output lower limit V_{RMIN}	pu	0
`ka`	Amplifier gain K_A	pu	1
`ta`	Amplifier time constant T_A	s	0
`te`	Exciter time constant T_E	s	0.4
`vfemax`	Exciter field-current limit V_{FEMAX}	pu	11.46
`vemin`	Minimum exciter voltage V_{EMIN}	pu	0
`ke`	Exciter self-excitation constant K_E	pu	1
`kc`	Rectifier loading factor K_C	pu	0.17
`kd`	Exciter demagnetising factor K_D	pu	1.19
`e1`	Exciter voltage at saturation point 1 E_1	pu	4.8
`se1`	Saturation at E_1, $S_E(E_1)$	–	0.0222
`e2`	Exciter voltage at saturation point 2 E_2	pu	3
`se2`	Saturation at E_2, $S_E(E_2)$	–	0.0219
`vtmult`	Terminal-voltage multiplier on regulator limits (0 = off)	–	0
`spdmlt`	Speed multiplier on exciter output (0 = off)	–	0

Table 8: `esac8b` parameters (common to all nine units).

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6 Turbine-governor — ggov1

ggov1 is the general-purpose GE governor and turbine model. It combines a PID speed governor with permanent droop, a load (temperature) limiter and an acceleration limiter, selects the lowest of the three demands, and drives a rate- and position-limited actuator into a simple turbine block, Figure 4.

With rselect $= 1$ the droop feedback is taken from electrical power through the $T_{pelec} = 1$ s transducer, giving a 5 % droop on the mwcap base (65 MW for the Turboden units, 60.461 MW for the Baker Hughes units — these are the only two parameters that differ between families). The speed error is clipped to ± 0.05 pu; the governor PI gains are $K_{Pgov} = 10$, $K_{Igov} = 2$ with the derivative path disabled. The actuator has $T_{act} = 0.5$ s, a valve range of 0 to 1 pu, an unlimited opening rate and a closing rate of -0.1 pu/s. The turbine is $P_{mech} = K_{turb} \cdot (F_{SR} - W_{fnl})$ passed through a $1/(1 + s \cdot T_b)$ lag with $T_b = 0.12$ s, $K_{turb} = 1.1$ and $W_{fnl} = 0.1$ pu; flag $= 0$ makes fuel flow independent of speed and teng $= 0$ removes the engine transport delay, both as expected for a steam turbine. The load limiter is set to $L_{dref} = 1.0$ with $K_{Pload} = 2$, $K_{Iload} = 0.67$ and $T_{fload} = 3$ s, the acceleration limiter to $a_{set} = 1$ pu/s with $K_a = 10$, $T_a = 0.1$ s. The supervisory MW controller is disabled ($K_{imw} = 0$) and mechanical damping is zero.

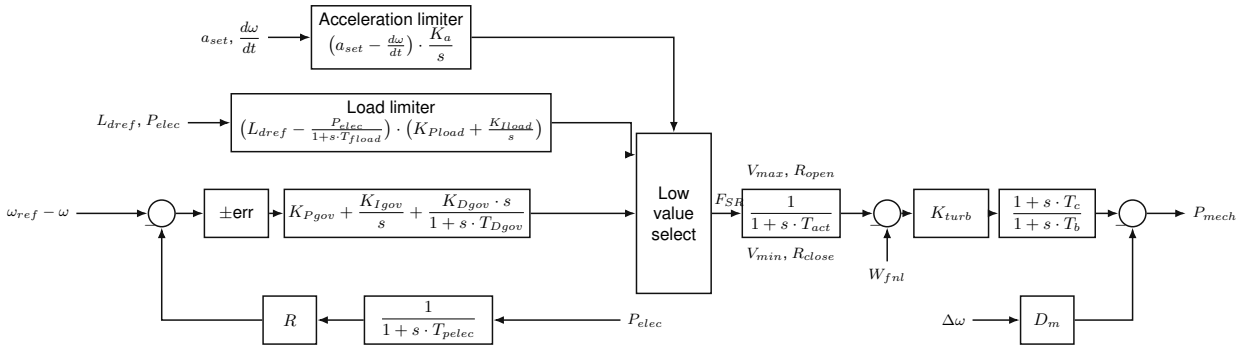


Figure 4: ggov1 block diagram (structure). With the present settings $K_{Dgov} = 0$, $T_c = 0$, $D_m = 0$ and $K_{imw} = 0$, so the derivative path, the turbine lead, the damping term and the MW controller are inactive.

Table 9 lists all 36 parameters. The temperature-detection lead-lag ($T_{sa} = 4$ s, $T_{sb} = 5$ s) and the load-limit ramp rates (± 99 pu/s) belong to the load-limiter path and are effectively non-binding.

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Parameter	Description	Unit	Turboden	Baker Hughes
mwcap	Turbine MW capability (governor base)	MW	65	60.461
r	Permanent droop R	pu	0.05	0.05
rselect	Feedback signal for droop (1 = electrical power)	–	1	1
tpelec	Electrical-power transducer time constant T_{pelec}	s	1	1
maxerr	Maximum speed-error signal	pu	0.05	0.05
minerr	Minimum speed-error signal	pu	-0.05	-0.05
kpgov	Governor proportional gain K_{Pgov}	pu	10	10
kigov	Governor integral gain K_{Igov}	pu/s	2	2
kdgov	Governor derivative gain K_{Dgov}	pu·s	0	0
tdgov	Governor derivative time constant T_{Dgov}	s	1	1
vmax	Maximum valve position V_{max}	pu	1	1
vmin	Minimum valve position V_{min}	pu	0	0
tact	Actuator time constant T_{act}	s	0.5	0.5
kturb	Turbine gain K_{turb}	pu	1.1	1.1
wfnl	No-load fuel flow W_{fnl}	pu	0.1	0.1
tb	Turbine lag time constant T_b	s	0.12	0.12
tc	Turbine lead time constant T_c	s	0	0
flag	Fuel-flow speed dependence (0 = independent of speed)	–	0	0
teng	Transport lag for diesel engine T_{eng}	s	0	0
tfload	Load-limiter transducer time constant T_{fload}	s	3	3
kpload	Load-limiter proportional gain K_{Pload}	pu	2	2
kiload	Load-limiter integral gain K_{Iload}	pu/s	0.67	0.67
ldref	Load-limiter reference L_{dref}	pu	1	1
dm	Mechanical damping / speed sensitivity D_m	pu	0	0
ropen	Maximum valve opening rate R_{open}	pu/s	0	0
rclose	Maximum valve closing rate R_{close}	pu/s	-0.1	-0.1
kimw	Power-controller (MW) integral gain K_{imw}	pu/s	0	0
pmwset	Power-controller setpoint P_{mwset}	MW	0	0
aset	Acceleration-limiter setpoint a_{set}	pu/s	1	1
ka	Acceleration-limiter gain K_a	pu	10	10
ta	Acceleration-limiter time constant T_a	s	0.1	0.1
db	Speed governor deadband	pu	0	0
tsa	Temperature-detection lead time constant T_{sa}	s	4	4
tsb	Temperature-detection lag time constant T_{sb}	s	5	5
rup	Maximum rate of load-limit increase	pu/s	99	99
rdown	Maximum rate of load-limit decrease	pu/s	-99	-99

Table 9: ggov1 parameters (pu on mwcap base unless noted).

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7 Power system stabilizer — pss2a

7.1 Purpose of a power system stabilizer

A power system stabilizer (PSS) is a supplementary control loop on the excitation system of a synchronous generator. Its only task is to damp the electromechanical oscillations of the rotor against the rest of the grid; it contributes nothing to the steady-state voltage or power setpoints and is inactive when the machine runs at constant speed.

The need for it arises from the voltage regulator itself. A modern static or brushless excitation system with a high-gain, fast-acting automatic voltage regulator (AVR) such as the esac8b of Section 5 holds the terminal voltage tightly and thereby raises the synchronising torque of the machine, which improves transient stability after a fault. The same fast field action, however, produces a component of electrical torque that lags the rotor speed deviation, which reduces — and in weakly connected systems can eliminate — the natural damping of the rotor swing. The result is a growing or poorly damped oscillation of rotor angle, power and voltage in the range of roughly 0.1 to 2 Hz: local plant modes near 1–2 Hz, in which a unit or a plant swings against the nearby system, and inter-area modes near 0.1–0.8 Hz, in which groups of machines in different regions swing against each other [3].

The PSS restores this damping. It measures a signal that reflects the rotor speed deviation and, after phase compensation, adds a small modulating signal V_S to the voltage reference of the AVR. The phase compensation is tuned so that the resulting change in electrical torque is in phase with the speed deviation; a torque component in phase with speed is by definition a damping torque. The output is limited (here to ± 0.1 pu) so that the stabilizer can only ever nudge the terminal voltage and cannot override the voltage regulator during a large disturbance.

Early stabilizers used speed alone as the input, which made them sensitive to torsional shaft modes, or electrical power alone, which caused an unwanted terminal-voltage excursion whenever the turbine output was ramped. The *integral-of-accelerating-power* design standardised as IEEE 421.5 type PSS2A [4] combines both: speed and electrical power are used to synthesise the integral of the accelerating power, $\int (P_{mech} - P_{elec})\,dt$, which is proportional to the speed deviation but free of the torsional content and insensitive to slow changes in mechanical power. This is the design applied to all nine Cape Station units.

For the Fervo units the stabilizer serves two purposes. It provides the damping that the fast esac8b regulator would otherwise remove, and it satisfies the WECC requirement that synchronous generators equipped with suitable excitation systems be fitted with a stabilizer and have it in service, which is a condition of model acceptance for the interconnection studies this data set feeds.

7.2 Model implementation

pss2a is the IEEE 421.5 dual-input stabilizer. Input 1 is rotor speed deviation ($\mathtt{j1} = 1$) and input 2 is electrical power ($\mathtt{j2} = 3$), both measured locally. The two signals are washed out ($T_{w1} = T_{w2} = T_{w3} = 2$ s, $T_{w4} = 0$ so the fourth washout is bypassed), the power signal is integrated through $K_{S2}/(1 + s \cdot T_7)$ with $T_7 = 2$ s and $K_{S2} = 0.501$, and the two are combined and passed through the ramp-tracking filter with $T_8 = 0.5$ s, $T_9 = 0.1$ s, $N = 1$, $M = 5$ to synthesise the integral of accelerating power. The result is amplified by $K_{S1} = 5$ and phase-compensated by two identical lead-lag stages $(1 + 0.25s)/(1 + 0.02s)$, then limited to ± 0.1 pu before it enters the exciter as V_S. Figure 5 shows the structure.

The value $K_{S2} = 0.501$ is consistent with the usual tuning rule $K_{S2} = T_7/(2 \cdot H)$ evaluated with $H = 2.0$ s, which does not match either machine inertia ($H = 1.49$ or 1.50 s would give $K_{S2} = 0.671$ or 0.667). This is worth confirming with the supplier of the dynamic data.

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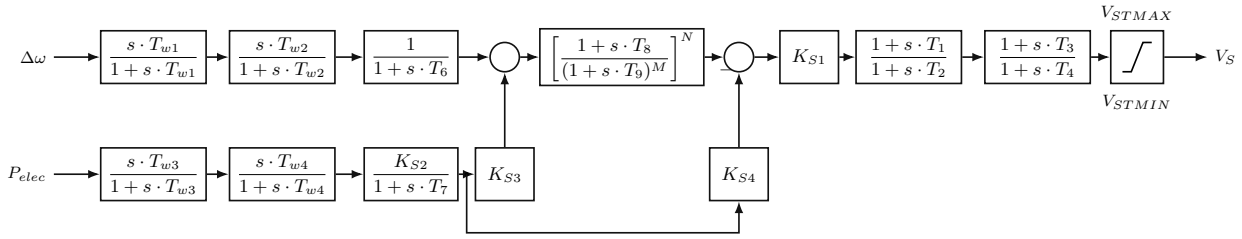


Figure 5: `pss2a` block diagram. $T_{w4} = 0$ bypasses the fourth washout; $K_{S3} = K_{S4} = 1$.

Parameter	Description	Unit	Value
`j1`	Input signal 1 code (1 = rotor speed deviation)	–	1
`k1`	Remote bus for input 1 (0 = local)	–	0
`j2`	Input signal 2 code (3 = electrical power)	–	3
`k2`	Remote bus for input 2 (0 = local)	–	0
`tw1`	Washout time constant T_{w1}	s	2
`tw2`	Washout time constant T_{w2}	s	2
`tw3`	Washout time constant T_{w3}	s	2
`tw4`	Washout time constant T_{w4} (0 = bypassed)	s	0
`t6`	Input 1 transducer time constant T_6	s	0
`t7`	Input 2 transducer time constant T_7	s	2
`ks2`	Input 2 gain K_{S2} ($= T_7/(2\cdot H)$)	pu	0.501
`ks3`	Signal-matching gain K_{S3}	pu	1
`ks4`	Input 2 gain K_{S4}	pu	1
`t8`	Ramp-tracking filter lead time constant T_8	s	0.5
`t9`	Ramp-tracking filter lag time constant T_9	s	0.1
`n`	Ramp-tracking filter order N	–	1
`m`	Ramp-tracking filter order M	–	5
`ks1`	Stabilizer gain K_{S1}	pu	5
`t1`	Lead time constant T_1	s	0.25
`t2`	Lag time constant T_2	s	0.02
`t3`	Lead time constant T_3	s	0.25
`t4`	Lag time constant T_4	s	0.02
`vstmax`	Stabilizer output upper limit V_{STMAX}	pu	0.1
`vstmin`	Stabilizer output lower limit V_{STMIN}	pu	-0.1
`a`	Optional second-order filter coefficient a	–	0
`ta`	Optional filter time constant T_a	s	0
`tb`	Optional filter time constant T_b	s	0

Table 10: `pss2a` parameters (common to all nine units).

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8 Ride-through protection — `lhvrt` and `lhfrt`

The `lhvrt` and `lhfrt` models are PSLF's generic low/high voltage and low/high frequency ride-through trip models. Each defines up to ten stages, a deviation from the reference (`vref` $= 1.0$ pu, `fref` $= 60$ Hz) and a time delay; when the monitored quantity stays beyond a stage's threshold for longer than that stage's delay, the unit is tripped. Both models are set to trip rather than alarm (`alarm` $= 0$) and are identical across all nine units.

Four voltage stages and eight frequency stages are populated, Tables 11 and 12; the remaining stages are zero and inactive. Figure 6 plots the resulting must-trip boundaries. The frequency envelope follows the familiar WECC off-nominal frequency shape: continuous operation between 59.4 and 60.6 Hz, 180 s at ± 0.6 Hz, 30 s at 58.4/61.6 Hz and instantaneous (50 ms) trip below 57.0 Hz or above 61.7 Hz.

Stage	ΔU	U	Time delay
	pu	pu	s
1	+0.25	1.25	0.5
2	+0.15	1.15	6
3	-0.15	0.85	7
4	-0.45	0.55	3

Table 11: `lhvrt` voltage trip stages (`vref` $= 1.0$ pu).

Stage	Δf	f	Time delay
	Hz	Hz	s
1	+1.70	61.7	0.05
2	+1.60	61.6	30
3	+0.60	60.6	180
4	-0.60	59.4	180
5	-1.60	58.4	30
6	-2.20	57.8	7.5
7	-2.70	57.3	0.75
8	-3.00	57.0	0.05

Table 12: `lhfrt` frequency trip stages (`fref` $= 60$ Hz).

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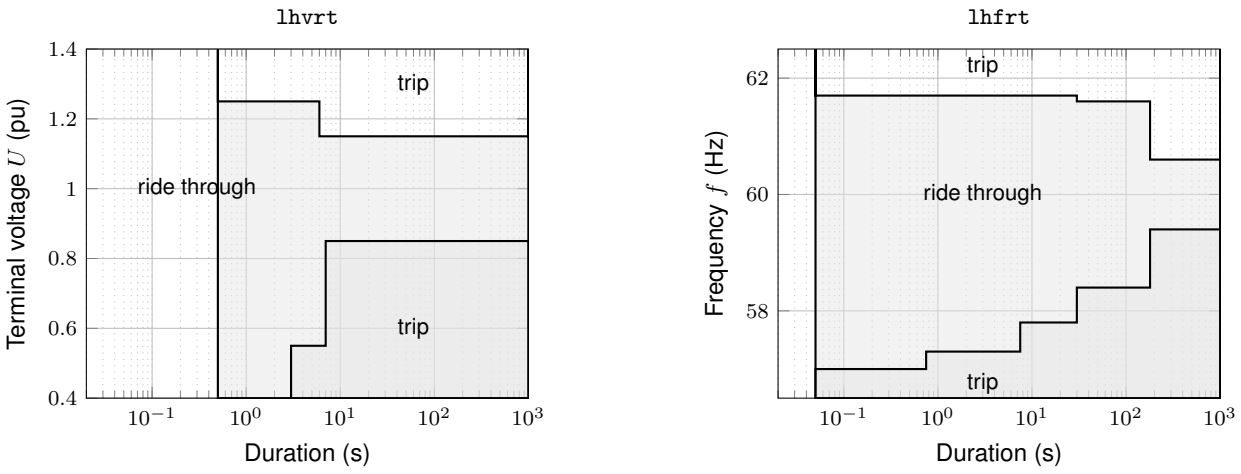


Figure 6: Must-trip regions (shaded) defined by the `lhvrt` and `lhfrt` settings. A unit trips once the terminal quantity has remained inside a shaded region for the duration on the abscissa.

9 Observations

A few points stand out from the two files and may deserve a check with the data provider before the model is used in studies.

The gross scheduled dispatch of 447 MW against 136 MW of auxiliary load yields about 311 MW net at the generator buses, well below the 400 MW in the case title, so the title figure refers to nameplate rather than to the dispatch in this case. Bus 22 hosts one Turboden and one Baker Hughes machine, so the two units on that bus have different dynamics even though they share a GSU. The GSU between buses 21 and 23 carries circuit identifier 11 where the other three carry 1, which is harmless but inconsistent. The `pss2a` gain $K_{S2} = 0.501$ implies a design inertia of $H = 2.0$ s rather than the modelled 1.49–1.50 s. Finally, the POI is modelled as an infinite bus (`gencls`, $H = 999$ s) and the Phase 1 tie is represented only by its line to bus 2 with no equipment behind it, so the case as delivered is a plant-only model intended to be merged into a full-system case for interconnection studies.

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10 References

[1] Fervo Energy, *Cape Station Phase 2 — PSLF power-flow network case*, filed with this report as `references/ModelEPC_Fervo_CapeStation_Ph2_v24_2026.09.05.epc`, PSLF version 24.3.0, exported September 5, 2026.

[2] Fervo Energy, *Cape Station Phase 2 — PSLF dynamic data*, filed with this report as `references/ModelDYD_Fervo_CapeStation_Ph2_2026.09.05.dyd`, September 5, 2026.

[3] P. Kundur, *Power System Stability and Control*, McGraw-Hill, 1994, chapter 12 (small-signal stability) and section 17.2 (power system stabilizers).

[4] IEEE Std 421.5-2016, *IEEE Recommended Practice for Excitation System Models for Power System Stability Studies*, IEEE Power and Energy Society, 2016. Type AC8B excitation system and type PSS2A stabilizer.

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